

Results — Soil Nitrogen Projection Model v2.1 (with SPEI)

Nine-plot rotational grazing / cover crop trial, 2019–2023. Response variables: NH_4^+ and NO_3^- (0–10 cm and 10–20 cm).

Figures 1 & 2 — Mean Trajectories and Distribution by Year

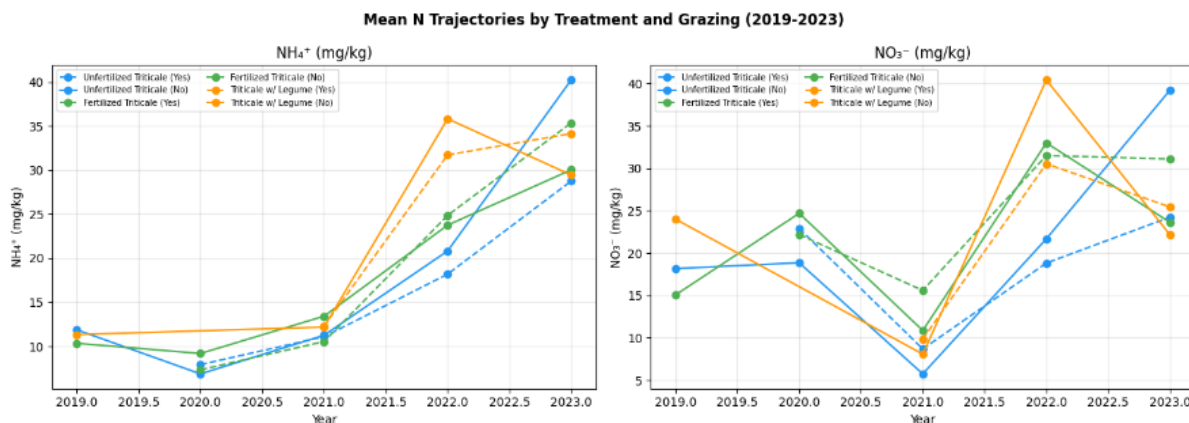


Figure 1. Mean NH_4^+ and NO_3^- trajectories by treatment, 2019–2023.

NH_4^+

All three treatments follow a similar trajectory: values remain low and comparatively stable from 2019 to 2021, followed by a marked increase in 2022–2023. Treatment-level differences are minimal across the series. This pattern is consistent with the increase being driven primarily by a system-wide factor — such as post-drought recovery, cumulative organic matter decomposition, or grazing timing — rather than by treatment-specific effects.

NO_3^-

Trajectories differ more by treatment. Triticale interseeded with Legume begins at the highest concentration in 2019 (approximately 24 mg/kg) before declining. All three treatments reach a low point in 2021, followed by an increase in 2022. The 2021 decline coincides with documented severe drought conditions in Nebraska, a period associated with reduced nitrification rates, consistent with a climate-driven signal rather than sampling noise.

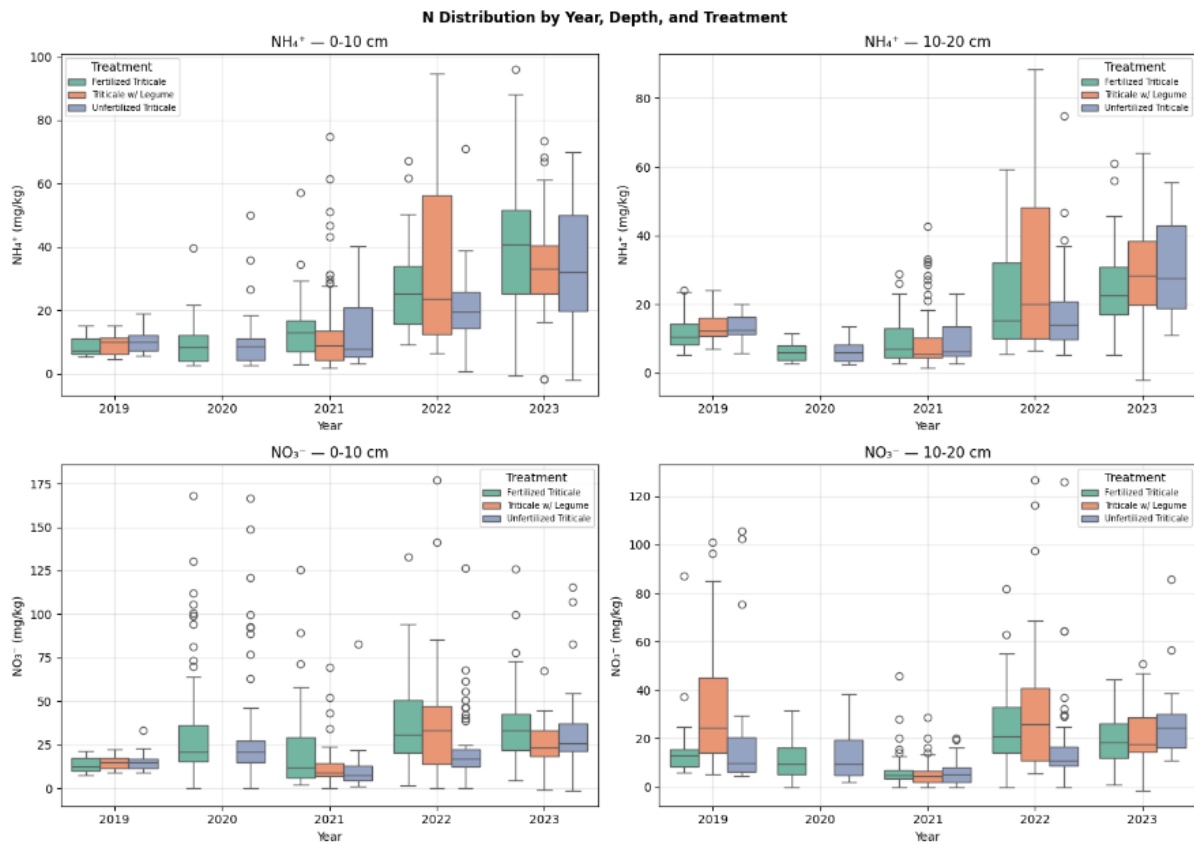


Figure 2. Boxplots of NH_4^+ and NO_3^- concentrations by year and depth.

Figure 2 — Boxplots

Three observations are notable:

- Within-treatment variability is high, particularly in 2022–2023. This is consistent with grazed and non-grazed subplots being aggregated within treatment groups at this stage of analysis; the width is reflected in the corresponding GPR uncertainty bands.
- NO_3^- shows extreme values in 2019–2020 (up to 170 mg/kg at 0–10 cm). These are consistent with localized inorganic nitrogen hotspots rather than measurement error, and contribute to elevated GPR mean estimates for those years.
- No boxes are present for the 10–20 cm depth in 2020, consistent with the previously identified 2020 depth-coverage gap in the dataset.

Figures 3 & 4 — Random Forest Feature Importance

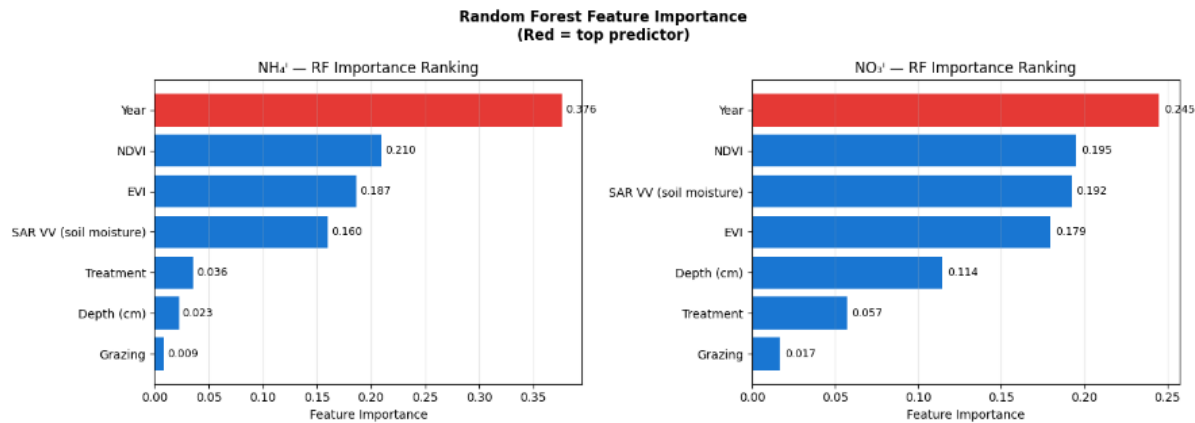


Figure 3. RF feature importance rankings, base model (without SPEI).

Feature importance (base model)

Feature	NH4+	NO3-	Interpretation
Year	0.376	0.245	Temporal trend dominates, consistent with a five-year rising trajectory.
NDVI	0.210	0.195	Vegetation cover is a strong proxy for N uptake and cycling.
EVI	0.187	0.179	Correlated with NDVI; both capture the biomass signal.
SAR VV	0.160	0.192	Reflects soil moisture, which is linked to nitrification rate.
Treatment	0.036	0.057	Low importance; treatments have not diverged substantially over five years.
Depth	0.023	0.114	More important for NO3-, consistent with downward leaching.
Grazing	0.009	0.017	Lowest importance; grazing is not a strong predictor of bulk N pool size.

Cross-validation results

Five-fold cross-validated R²: −0.198 (NH4+) and −0.199 (NO3-), indicating that RF predictions underperform relative to predicting the mean. This is consistent with the limited number of years (5) relative to the number of folds, leaving approximately 3–4 training years per fold.

Leave-one-year-out cross-validation was used as an alternative:

NH4+ RF results

Cross-validated R² (5-fold): −2.592

Feature importances: year 0.3765, ndvi 0.2097, evi 0.1869, sar_vv 0.1604, treatment_code 0.0355, depth_numeric 0.0225, grazing_binary 0.0085

Leave-one-year-out R^2 scores: -1.377 , -8.379 , -0.010 , -1.482 , -1.711

Mean leave-one-year-out R^2 : -2.592

NO3- RF results

Cross-validated R^2 (5-fold): -0.868

Feature importances: year 0.2451, ndvi 0.1951, sar_vv 0.1922, evi 0.1793, depth_numeric 0.1145, treatment_code 0.0570, grazing_binary 0.0168

Leave-one-year-out R^2 scores: -0.702 , -0.130 , -2.031 , -1.092 , -0.385

Mean leave-one-year-out R^2 : -0.868

Interpretation of negative R^2

All leave-one-year-out folds return negative R^2 , including the best-performing fold (-0.01 , NH4+, likely 2021). This indicates that RF does not predict nitrogen values in an unseen year more accurately than the training-set mean. The lowest-performing fold (-8.379) corresponds to the 2021 drought year: when 2021 is held out as the test set, no comparable anomaly is present in the training years.

This outcome reflects a structural limitation of the dataset — five annual time points — rather than an implementation error. RF cross-validation is expected to perform poorly with this few training examples per fold regardless of model configuration.

Role of RF in the pipeline

Role	Valid	Basis
Predictive model	No	Negative R^2 indicates the model does not predict unseen years.
Feature selection / ranking tool	Yes	Importance rankings are stable and interpretable independent of R^2 .
Covariate prioritization for GPR	Yes	Identifies which covariates to weight; GPR handles uncertainty explicitly.

Recommended framing for methods/abstract

“Random Forest feature importance ranking was used to prioritize covariates for GPR, given the limited temporal replication inherent in a five-year observational dataset.”

Suggested methods note: “Due to the five-year temporal extent of the dataset, leave-one-year-out cross-validation yielded negative R^2 values, indicating that RF does not generalize to unseen years at this sample size. RF importances were therefore used for covariate prioritization rather than as a standalone predictive model.”

Recommendation: report feature importance rankings rather than R^2 values in presentation materials.

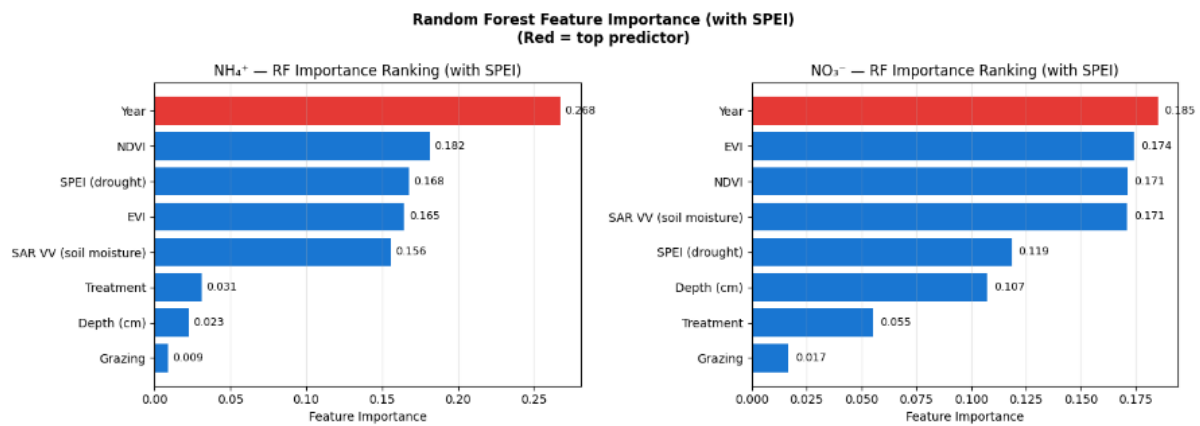


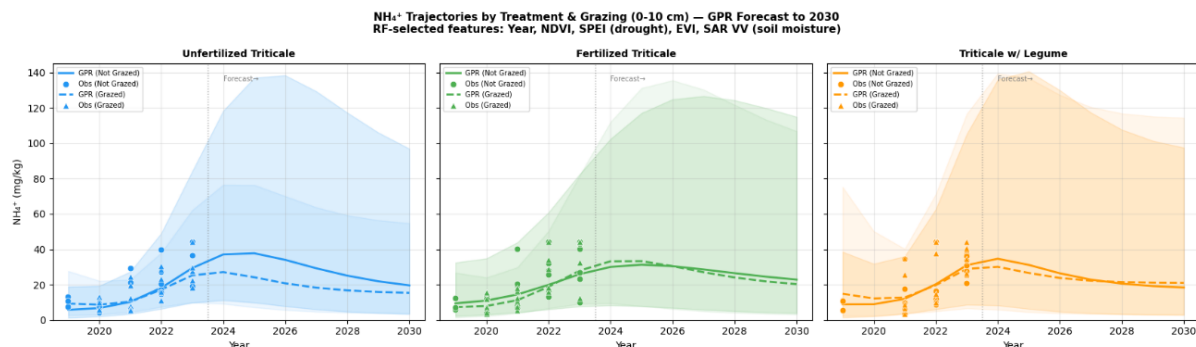
Figure 4. RF feature importance rankings with SPEI included.

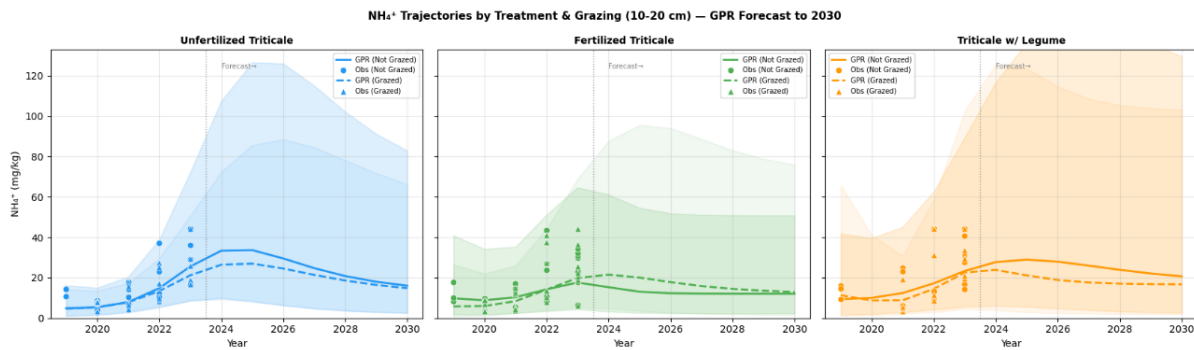
Feature importance with SPEI

Feature	NH ₄ ⁺ rank	NO ₃ ⁻ rank
Year	1st (0.268)	1st (0.185)
NDVI	2nd	3rd
SPEI	3rd (0.168)	5th (0.119)
EVI	4th	2nd
SAR VV	5th	4th
Treatment	6th	7th
Depth	7th	6th
Grazing	Last (0.009)	Last (0.017)

SPEI ranks third for NH₄⁺ and fifth for NO₃⁻, indicating that drought conditions provide explanatory value for ammonium dynamics beyond noise. Grazing ranks last for both response variables, consistent with a small effect size at the bulk pool level.

Figures 5 & 6 — NH₄⁺ Forecasts by Depth and Grazing Status





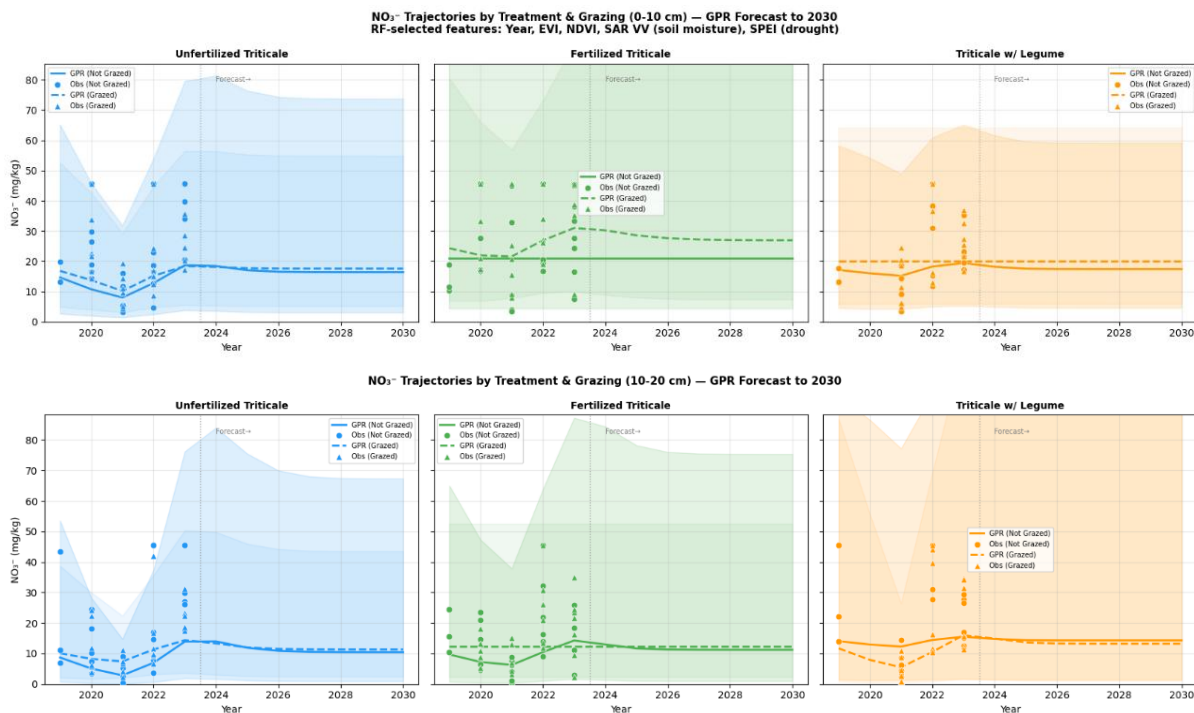
Figures 5–6. GPR forecasts for NH₄⁺ at 0–10 cm and 10–20 cm, grazed vs. not grazed.

A consistent pattern is observed across both depths:

- Non-grazed plots forecast higher NH₄⁺ than grazed plots, particularly within Unfertilized Triticale, consistent with ammonium removal by grazing exceeding accumulation in the absence of an external N input.
- Fertilized Triticale shows the smallest difference between grazed and non-grazed forecasts, consistent with fertilizer N input offsetting removal by grazing.
- Triticale with Legume shows near-overlapping grazed/non-grazed trajectories, consistent with legume N fixation compensating for grazing removal.

Confidence intervals remain wide for both NH₄⁺ and NO₃⁻ following the addition of SPEI, reflecting the limited number of plot-level observations per year rather than a modeling deficiency. Band width is comparable between the two response variables. For presentation purposes, mean trajectories without confidence bands are recommended for both NH₄⁺ and NO₃⁻.

Figures 7 & 8 — NO₃⁻ Forecasts by Depth and Grazing Status



Figures 7–8. GPR forecasts for NO₃⁻ at 0–10 cm and 10–20 cm, grazed vs. not grazed.

0–10 cm

- Fertilized Triticale: grazed plots forecast higher NO₃⁻ than non-grazed plots, consistent with grazing accelerating N cycling through dung inputs undergoing nitrification.
- Unfertilized Triticale: grazed and non-grazed trajectories converge, consistent with grazing having minimal net effect on NO₃⁻ at this depth in the absence of an external N input.
- Triticale with Legume: grazed and non-grazed trajectories are nearly identical, consistent with legume N fixation dominating over grazing effects.

10–20 cm

- Concentrations are lower overall across treatments, consistent with downward leaching of NO₃⁻.
- Grazed Unfertilized plots trend higher than non-grazed plots at this depth, consistent with greater accumulation of leachable NO₃⁻ under grazing.

Figure 9 — Grazing Effect Heatmap

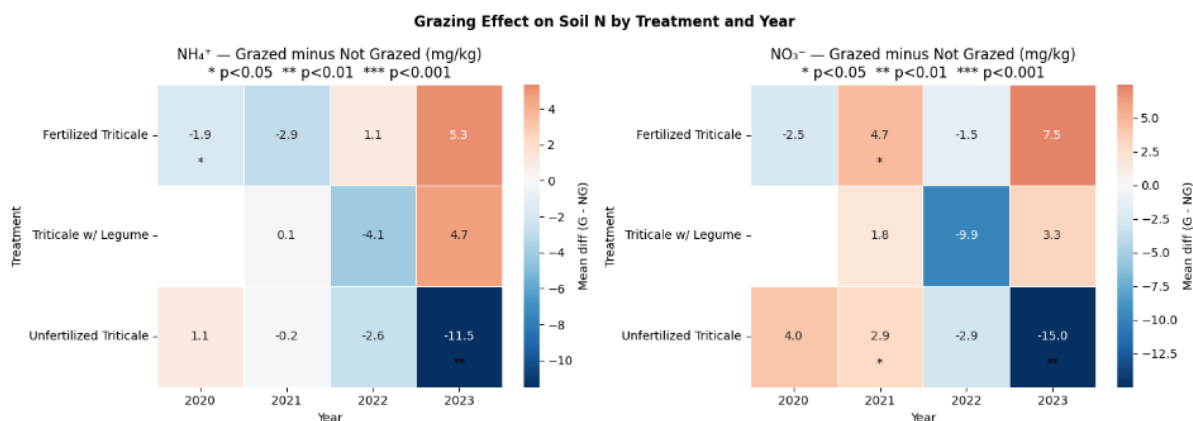


Figure 9. Estimated grazing effect on NH₄⁺ and NO₃⁻ by treatment and year.

NH₄⁺

- Unfertilized, 2023: -11.5 mg/kg (p < 0.01) — statistically significant depletion of ammonium under grazing in plots without fertilizer or legume input.
- Fertilized, 2023: +5.3 mg/kg (not significant) — a modest increase associated with grazing, consistent with dung deposition.
- Triticale with Legume, 2023: +4.7 mg/kg (not significant) — a comparable non-significant increase.

NO₃⁻

- Unfertilized, 2023: -15.0 mg/kg (p < 0.01) — significant depletion, larger in magnitude than the corresponding NH₄⁺ effect.
- Triticale with Legume, 2021: -9.9 mg/kg — a large negative effect during the drought year, potentially reflecting reduced legume N fixation under drought stress combined with continued grazing pressure.
- Fertilized, 2021: +4.7 mg/kg (p < 0.05) — grazing associated with increased N mineralization in fertilized plots under drought conditions.

Summary

The grazing effect on soil N pools depends on the presence of an external N input. In plots receiving fertilizer or legume-associated N inputs, the effect of grazing on NH_4^+ and NO_3^- is neutral to slightly positive. In unfertilized plots, grazing is associated with statistically significant depletion of both NH_4^+ and NO_3^- , an effect that increases in magnitude through 2023.

Model v2.1 — Overall Summary

Three findings are highlighted from this analysis:

- Year and remote sensing indices (NDVI, EVI, SAR VV) are the strongest predictors of soil N dynamics; treatment alone accounts for a small proportion of variance over the five-year period.
- The SPEI drought index captures the 2021 decline in nitrogen concentrations, indicating climate variability as a confounding factor in treatment comparisons.
- The grazing $\times$ treatment interaction is the most consistent biological signal in the dataset: grazing is associated with significant soil N depletion in unfertilized plots ($p < 0.01$), while remaining neutral to slightly positive in fertilized and legume-interseeded plots.